

Energy-Aware FPGA Implementation of Spiking Neural Network with LIF Neurons

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Preface & Motivation

Spiking Neural Networks (SNNs) represent input intensities as randomized sequences of 0s and 1s over several time steps, where the frequency of 1s is proportional to the input intensity. Because each input is binary, multiplication can be replaced with MUXes that select either the corresponding weight or zero, followed by simpler adder-based processing. This makes SNNs well suited to vision-based collision avoidance on tiny UAVs/UGVs, where real-time inference must operate under strict power, memory, and hardware constraints. Compared with conventional neural networks built around multi-bit MAC operations, SNNs can therefore reduce FPGA energy consumption.

Problem Statement

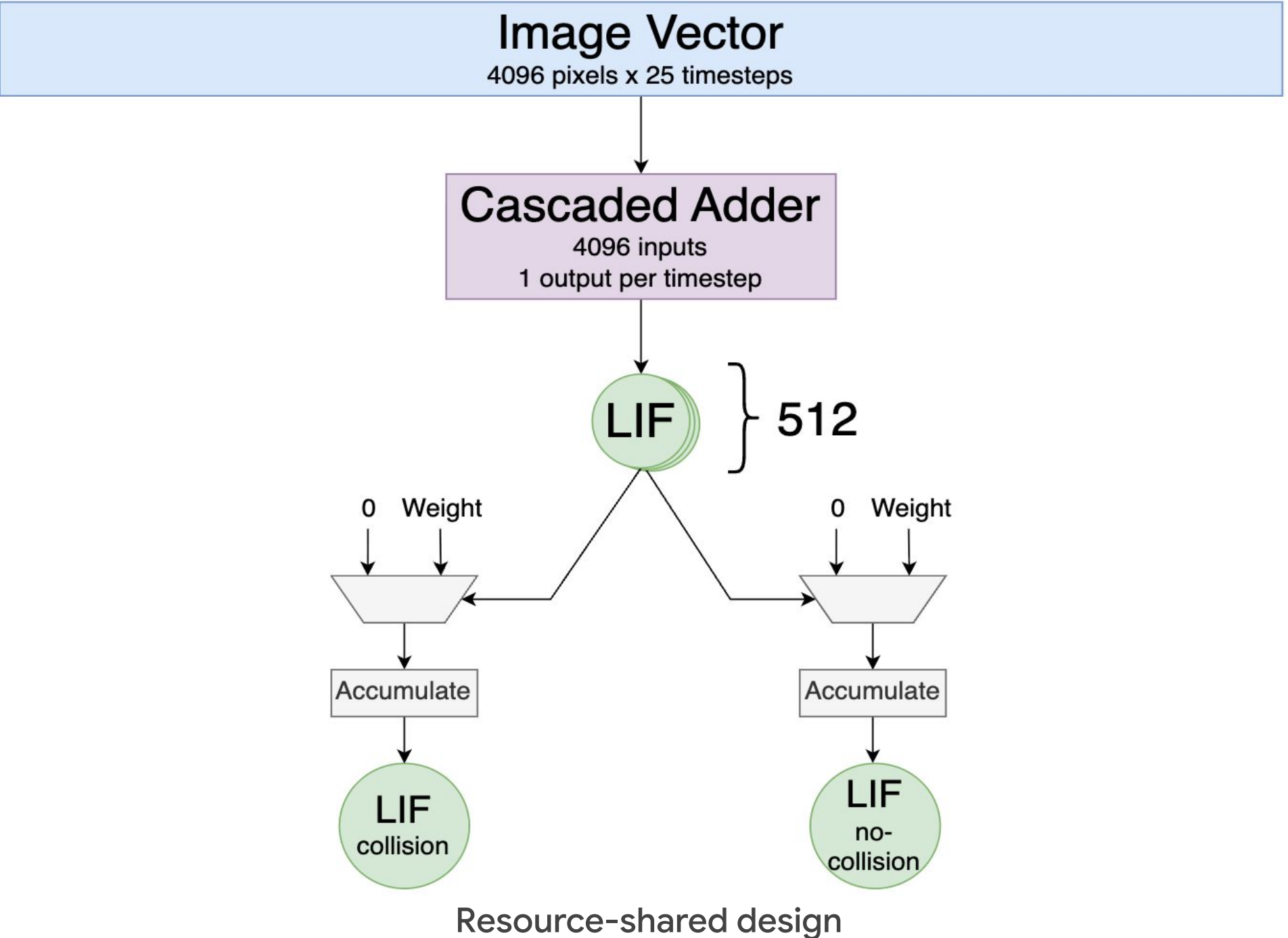
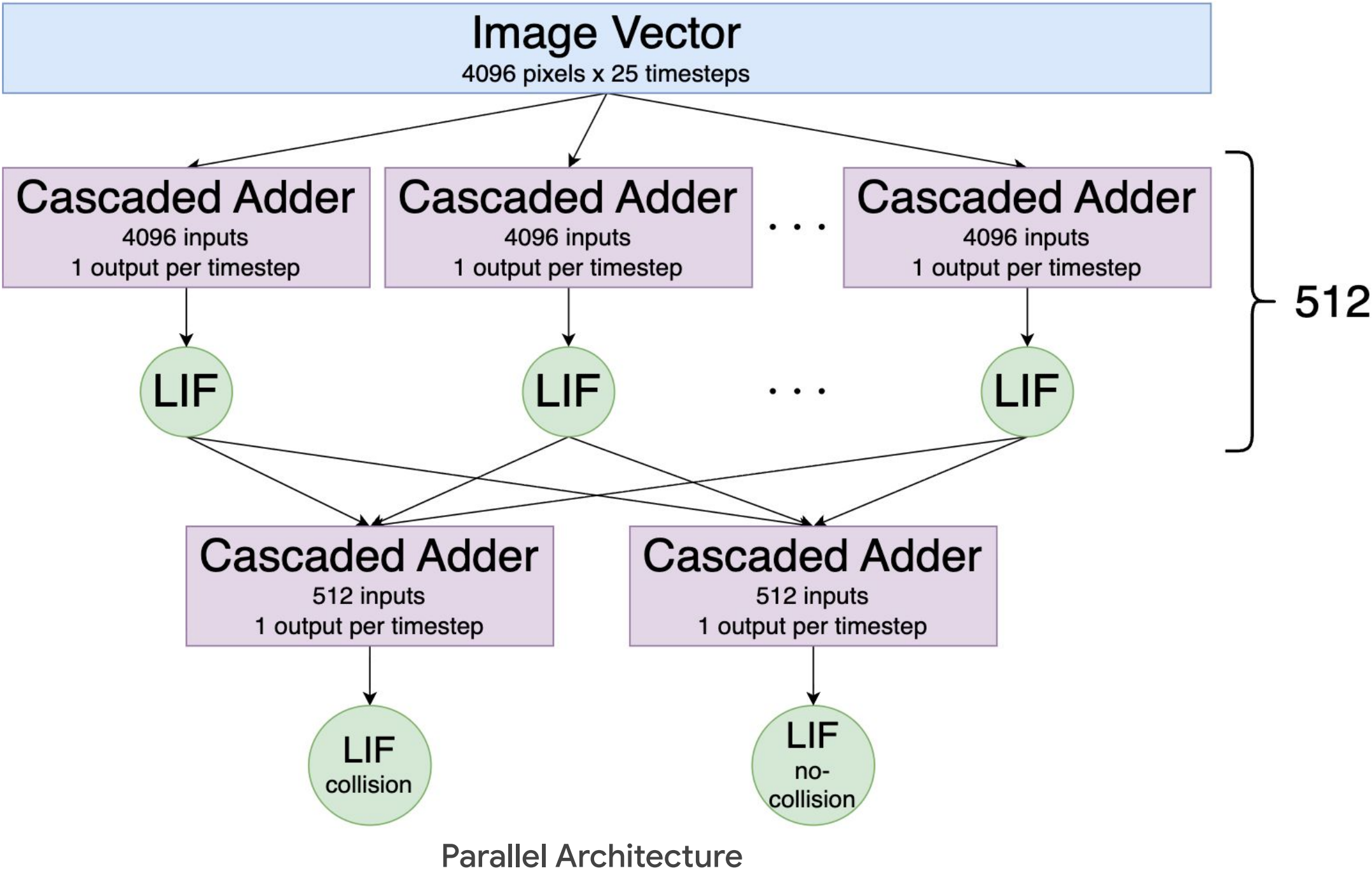
We set out to independently evaluate the paper’s central claim that an FPGA-based SNN can achieve both lower power consumption and higher throughput than conventional neural network accelerators (specifically the BCNN baseline specified in the paper).

Paper Citation: Asmer Hamid Ali, Mozghan Navardi, and Tinoosh Mohsenin, “Energy-Aware FPGA Implementation of Spiking Neural Network with LIF Neurons,” arXiv:2411.01628, 2024

Project Scope

We attempted to replicate the paper’s 4096-512-2 SNN model in RTL using 2 microarchitectures:

- (1) Area-oriented design: one 4096-input cascaded adder is reused across the 512 hidden neurons, and hidden spikes are accumulated into the two output LIF neurons.
(2) Throughput-oriented parallel design: 512 parallel 4096-input adder trees and LIF neurons feed two 512-input output adder trees and LIF neurons.
Weights and biases are preloaded from software exported hex tables; the testbench streams offline encoded spike vectors. Functionally verified in simulations, not deployed on FPGA.



The LIF Neuron Unit

The Leaky-Integrate-and-Fire (LIF) neuron model is a central architecture piece of our design. Every time step (i.e., every set of randomly generated input spikes), the LIF neuron loses a small percentage of its current value, but gains some new amount of input current, equal to the sum of the weights from the active spikes in the previous layer.

Once the internal value of the LIF exceeds the threshold value, the LIF must output a spike. At this point, the LIF internal value is reset to zero, and further spikes are suppressed for 5 timesteps. Internal value decay rate values and input current bias values were learnable parameters through software. The hardware and software teams arrived at the following architecture for the LIF model:

- The LIF spikes on the same time step as the internal value surpasses the threshold value. This allows for the possibility that the output spikes even as soon as the first time step.
- The LIF internal value continues to accumulate as normal during the 5-time-step cooldown period, only suppressing spikes. This was our selected interpretation of the paper’s *underspecified* refractory behavior.

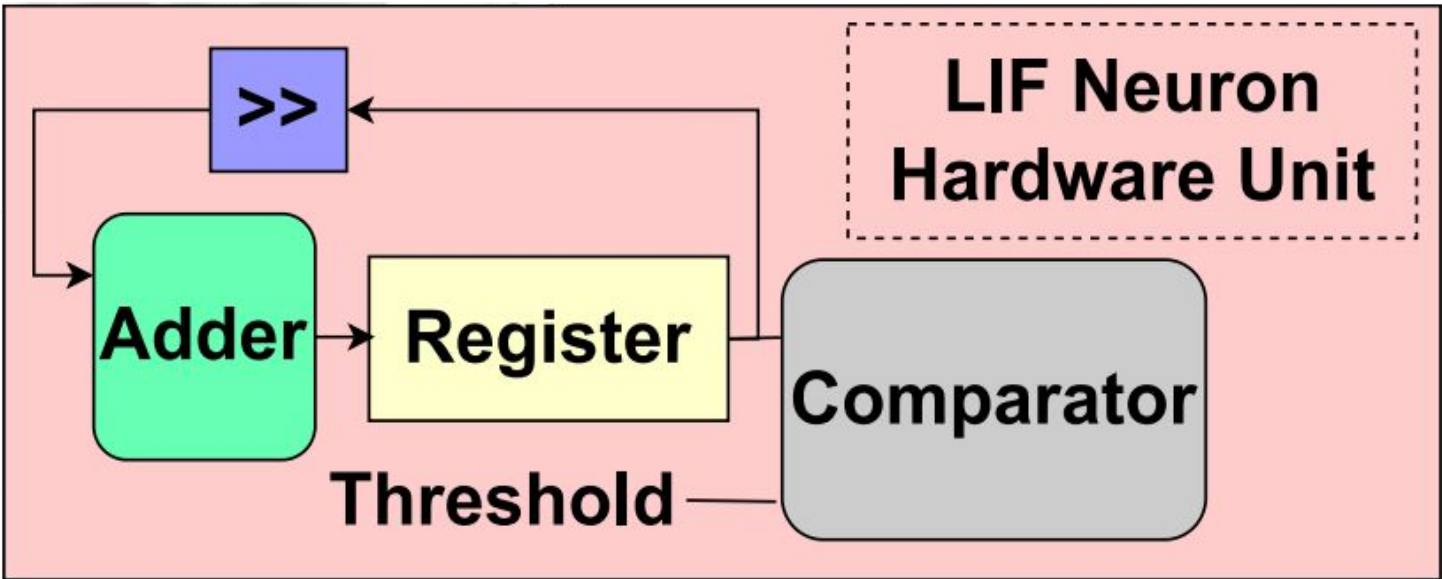


Figure illustrating the proposed flow of hardware implementation for a Spiking Neural Network (Ali et al., Fig. 5.)

Preliminary Results

Result	Paper-Reported SNN	BCNN baseline	Our design @67 MHz	Our design @280 MHz
Hardware scope	4096-512-2 SNN	VGG-11 BCNN	One 4096-input tree + one LIF	One 4096-input tree + one LIF
FPGA	Artix-7	Zynq-7000	AU25P UltraScale+	AU25P UltraScale+
Frequency	67 MHz	143 MHz	67 MHz	280 MHz
Power	0.495 W	2.3 W	1.332 W total*, 0.875 W dynamic	3.874 W total*, 3.395 W dynamic
Peak GOPS	541	329	548.88 (peak synaptic arith rate)	2294
Peak GOPS/W	1093	143	412	592
LUTs	6521	Not given	38667	38713
Registers/FFs	1780	Not given	69606	69606

Comparison of SNN with Paper & Previous Works

* Post-route estimates from Vivado 2022.1 with dynamic power SAIF annotated at 100% net matching

Verification

We verified the RTL in simulation with cocotb/Verilator. Below we cover more specifics:

- Unit Testing:** Cascaded adder & LIF neuron units
 - Cascaded adder was tested against software generated inputs/outputs: random/signed vectors, pipeline latency/throughput, reset.
 - LIF neuron was tested against a reference software model (Python reference): reset, enable, threshold fire, refractory, randomized lockstep + e2e collision predictor tests that check collision counters against test vectors.
- Integration Testing:**
 - Reference model:** Software model written with PyTorch and SNNtorch.
 - Inputs:** Images are taken from the Dronet collision dataset, converted to greyscale and resized to 64x64 pixels. There are 1576 images used for testing (validation split).
 - Timing/latency test:** 3 images are provided back to back and the latency of out_valid signals are verified.
 - Correctness test:** All 1576 images are streamed and the outputs are verified against the software model. Accelerator's number of "collision" and "no-collision" spikes should match exactly with the software model instead of just the plain binary output.
 - Throughput test:** For the cascaded design, images are streamed to fill up the pipeline 2 times over and out_valid signal frequency and amount are verified. For the low-area design, this test is a subset of the timing test.
 - All inclusive test:** Latency, throughput, and correctness are verified in a single test, with back-to-back inputting to fill up the pipeline 2 times over.

Model	Registers/FFs	LUTs	Frequency (MHz)	Power (mW)	Device
Izhikevich [1]	31	771	–	1043	Virtex-5
1st Order LIF (Paper)	96	25	100	85+	Artix-7
1st Order LIF (Ours)	32	155	100	6**	AU25P

Peak Compute-Unit Study vs. Reported Accelerator Results

* Paper does not specify whether its 85 mW value represents dynamic, total-device or measured power. The values are therefore not directly comparable. Differences may also reflect FPGA generation, implementation structure, switching activity and included clock/I/O resources.
** Post-route SAIF dynamic estimate including BUFG clock buffer, boundary registers and I/O buffers; estimated total on-chip power is 456 mW.

Note:
A reasonable explanation for the low power consumption in our study is the use of 16 nm transistors (AU25P) vs 28 nm (Artix-7). Note that the formula for dynamic power consumption is $P = aCV^2f$.

If the transistor size is scaled by 7/4 when we use a 28 nm architecture, the voltage will need to increase by a factor of 7/4 also, as would the capacitance. The result is that dynamic power losses would be scaled by >5x, which implies a power consumption in the ~30-40 mW range.

Next Steps...

- Reduce Register/FF and LUT counts by simplifying LIF design such that only the internal value and cooldown counter are saved between timesteps, and by streaming in weights from BRAM rather than saving them in registers.
- A more accurate test could be done with a 28 nm fabric, which would likely have less static power loss due to higher resistance ($P = V^2/R$).
- Time-multiplexing: one adder tree walks through all 512 neurons over (512/m) clocks per timestep, enabling 'm' LIFs at a time. Folded design for lower area.

[1] Shanmukha Murali, Juneeth Kumar, Jayanth Kumar, and Ramesh Bhakthavatchalu. 2016. Design and implementation of Izhikevich spiking neuron model on FPGA. 946–951. <https://doi.org/10.1109/RTEICT.2016.7807968>